

Test case in program

1.check the date in the range

Test case	Test case scenario	Inputs	outcomes	Remark
1	Check Day	33	Out of range-value must be in the range 1 and 31	pass
2	Check month	13	Out of range-value must be in the range 1 and 12	pass
3	Check year	2025	Out of range-value must be in the range 2000 and 2024	pass

Test case 1,2 and 3:

```
Please enter the day of the survey in the format DD: 33
Out of range - values must be in the range 1 and 31.
Please enter the day of the survey in the format DD: 16
please enter the month of the survey in the format MM: 13
Out of range - values must be in the range 1 and 12.
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2025
Out of range - values must be in the range 2000 and 2024.
```

2.Check the integer required error

Test case	Test case scenario	Inputs	outcomes	Remark
4	Check Day	ab	Integer required	pass
5	Check month	cd	Integer required	pass
6	Check year	de	Integer required	pass

Test case 4,5 and 6:

```
Please enter the day of the survey in the format DD: ab
Integer required
Please enter the day of the survey in the format DD: 15
please enter the month of the survey in the format MM: cd
Integer required
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: de
Integer required.
Please enter the year of the survey in the format YYYY: 2000
```

3.leap year, days of a month error

Test case	Test case scenario	Inputs	outcomes	Remark
7	Non-Leap year	29 02 2023	February has only 28 days in a non-leap year. Invalid data entered. please try again.	pass
8	leap year	30 02 2024	February has only 29 days in a leap year. Invalid data entered. please try again.	pass
9	Days of a month (30 or 31)	31 04 2024	This month has only 30 days. Invalid data entered. Please try again.	pass

Test case 7,8 and 9 :

```
Please enter the day of the survey in the format DD: 29
please enter the month of the survey in the format MM: 02
Please enter the year of the survey in the format YYYY: 2023
Februrary has only 28 days in a non-leap year
Invalid data entered.please try again.
Please enter the day of the survey in the format DD: 31
please enter the month of the survey in the format MM: 04
Please enter the year of the survey in the format YYYY: 2024
This month has only 30 days.
Invalid data entered.please try again.
Please enter the day of the survey in the format DD: 30
please enter the month of the survey in the format MM: 02
Please enter the year of the survey in the format YYYY: 2024
February has only 29 days in a leap year
Invalid data entered.please try again.
```

4.Validate continue input

Test case	Test case scenario	Inputs	outcomes	Remark
10	Validate continue input	abc	Invalid input. please enter 'Y' or 'N'	pass
11	End the program	n	End of run	pass

Test case 10 and 11 :

```
*****
Do you want to select a data file for a different date? (Y/N): abc
Invalid input. Please enter 'Y' or 'N'.
Do you want to select a data file for a different date? (Y/N): n
End of run.
*****
```

5.File not found error message

Test case	Test case scenario	Inputs	outcomes	Remark
12	If file isn't in the storage	23 06 2024	File not found. Please try again with a valid file	pass

Test case 12 :

```
Please enter the day of the survey in the format DD: 23
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
File not found.
Please try again with a valid file
```

6. Process outcomes and print

Test case	Test case scenario	Inputs	outcomes	Remark
13	Print outcomes for 15 th 06 2024	15 06 2024	Print all outcomes	pass
14	Print outcomes for 16 th 06 2024	16 06 2024	Print all outcomes	pass
15	Print outcomes for 21 st 06 2024	21 06 2024	Print all outcomes	pass

Test case 13 :

```
Please enter the day of the survey in the format DD: 15
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
*****
data file selected is traffic_data15062024.csv
The total number of vehicles recorded for this day is 1037
The total number of Trucks recorded for this day is 109
The total number of Electric vehicles recorded for this day is 368
The total number of two wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 15
The total number of vehicles passing through both junctions without turning left or right is 363
The percentage of total vehicles recorded that are Trucks for this date is 11%
The average number Bicycles per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through only Hanley Highway/Westway junction is 543
11% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded Between 18:00 and 19:00
The number of hours of rain for this date is 0

Processed data has been successfully saved to 'results.txt'
*****
Do you want to select a data file for a different date? (Y/N): y|
```

Test case 14 :

```
Please enter the day of the survey in the format DD: 16
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
*****
data file selected is traffic_data16062024.csv
The total number of vehicles recorded for this day is 101
The total number of Trucks recorded for this day is 11
The total number of Electric vehicles recorded for this day is 29
The total number of two wheeled vehicles for this date is 29
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 0
The total number of vehicles passing through both junctions without turning left or right is 38
The percentage of total vehicles recorded that are Trucks for this date is 11%
The average number Bicycles per hour for this date is 0
The total number of vehicles recorded as over the speed limit for this date is 20
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 52
The total number of vehicles recorded through only Hanley Highway/Westway junction is 49
11% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 5
The most vehicles through Hanley Highway/Westway were recorded Between 01:00 and 02:00
The number of hours of rain for this date is 3

Processed data has been successfully saved to 'results.txt'
*****
```

Test case 15 :

```
Please enter the day of the survey in the format DD: 21
please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024
*****
data file selected is traffic_data21062024.csv
The total number of vehicles recorded for this day is 1334
The total number of Trucks recorded for this day is 138
The total number of Electric vehicles recorded for this day is 442
The total number of two wheeled vehicles for this date is 503
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 19
The total number of vehicles passing through both junctions without turning left or right is 494
The percentage of total vehicles recorded that are Trucks for this date is 10%
The average number Bicycles per hour for this date is 10
The total number of vehicles recorded as over the speed limit for this date is 250
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 651
The total number of vehicles recorded through only Hanley Highway/Westway junction is 683
9% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 71
The most vehicles through Hanley Highway/Westway were recorded Between 18:00 and 19:00
The number of hours of rain for this date is 6

Processed data has been successfully saved to 'results.txt'
*****
Do you want to select a data file for a different date? (Y/N): n
End of run.
*****
```

Test case in program(task D/E)

1.check about outcomes and histogram for given csv files

Test case	Test case scenario	Inputs	outcomes	Remark
01	Check the histogram and outcomes for 2024/06/15	15 06 2024	Print the outcomes and display the histogram	pass
02	Check the histogram and outcomes for 2024/06/16	16 06 2024	Print the outcomes and display the histogram	pass
03	Check the histogram and outcomes for 2024/06/21	21 06 2024	Print the outcomes and display the histogram	pass

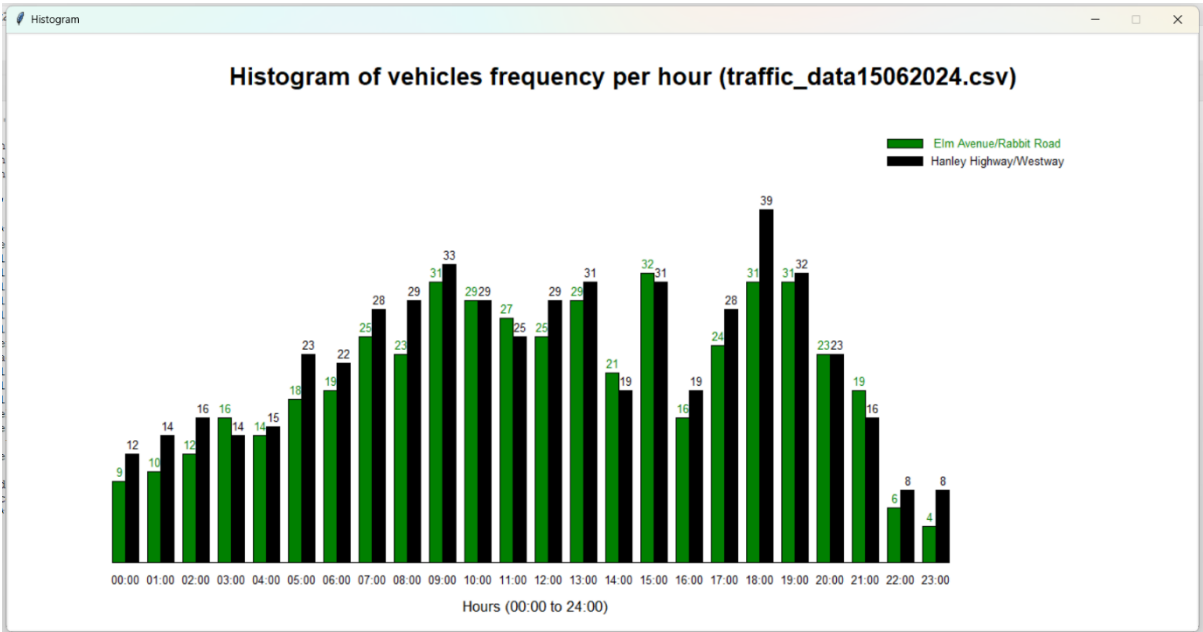
Test case 01:

```
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024

---File 'traffic_data15062024.csv' loaded successfully---
```

```
*****
data file selected is traffic_data15062024.csv
The total number of vehicles recorded for this day is 1037
The total number of Trucks recorded for this day is 109
The total number of Electric vehicles recorded for this day is 368
The total number of two wheeled vehicles for this date is 401
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 15
The total number of vehicles passing through both junctions without turning left or right is 363
The percentage of total vehicles recorded that are Trucks for this date is 11%
The average number Bicycles per hour for this date is 7
The total number of vehicles recorded as over the speed limit for this date is 205
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 494
The total number of vehicles recorded through only Hanley Highway/Westway junction is 543
11% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 39
The most vehicles through Hanley Highway/Westway were recorded Between 18:00 and 19:00
The number of hours of rain for this date is 0

Processed data has been successfully saved to 'results.txt'
click once on histogram if you want to select a data file for a different date.
*****
```



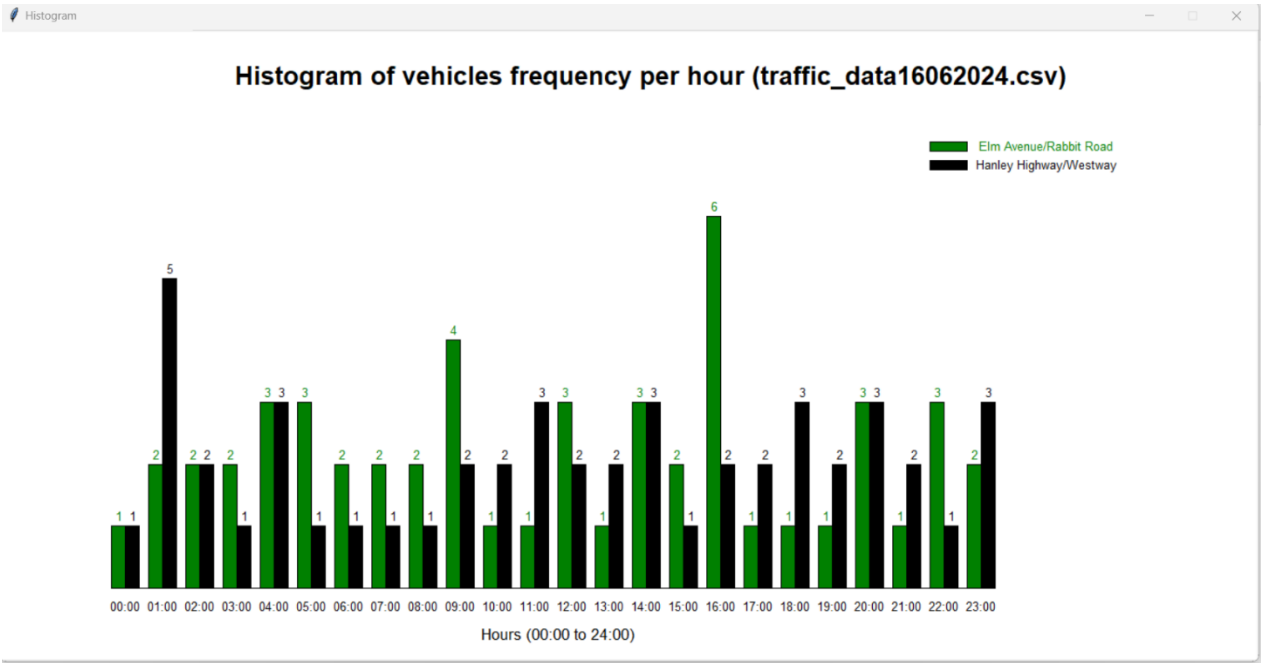
Test case 02:

```
Please enter the day of the survey in the format DD: 16
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024

---File 'traffic_data16062024.csv' loaded successfully---

*****
data file selected is traffic_data16062024.csv
The total number of vehicles recorded for this day is 101
The total number of Trucks recorded for this day is 11
The total number of Electric vehicles recorded for this day is 29
The total number of two wheeled vehicles for this date is 29
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 0
The total number of vehicles passing through both junctions without turning left or right is 38
The percentage of total vehicles recorded that are Trucks for this date is 11%
The average number Bicycles per hour for this date is 0
The total number of vehicles recorded as over the speed limit for this date is 20
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 52
The total number of vehicles recorded through only Hanley Highway/Westway junction is 49
11% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 5
The most vehicles through Hanley Highway/Westway were recorded Between 01:00 and 02:00
The number of hours of rain for this date is 3

Processed data has been successfully saved to 'results.txt'
click once on histogram if you want to select a data file for a different date.
*****
```



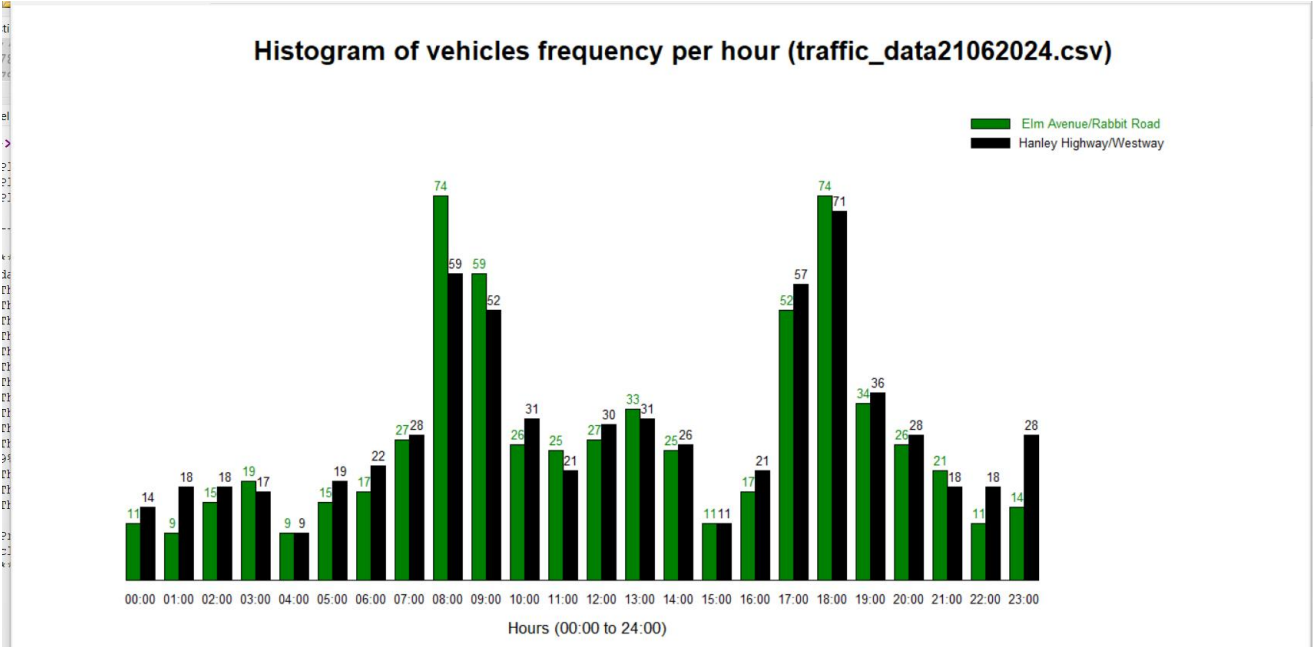
Test case 03:

```
Please enter the day of the survey in the format DD: 21
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024

---File 'traffic_data21062024.csv' loaded successfully---
```

```
*****
data file selected is traffic_data21062024.csv
The total number of vehicles recorded for this day is 1334
The total number of Trucks recorded for this day is 138
The total number of Electric vehicles recorded for this day is 442
The total number of two wheeled vehicles for this date is 503
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 19
The total number of vehicles passing through both junctions without turning left or right is 494
The percentage of total vehicles recorded that are Trucks for this date is 10%
The average number Bicycles per hour for this date is 10
The total number of vehicles recorded as over the speed limit for this date is 250
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 651
The total number of vehicles recorded through only Hanley Highway/Westway junction is 683
9% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 71
The most vehicles through Hanley Highway/Westway were recorded Between 18:00 and 19:00
The number of hours of rain for this date is 6

Processed data has been successfully saved to 'results.txt'
click once on histogram  if you want to select a data file for a different date.
*****
```



2.create a new csv data file and check the code run correctly.

Test case	Test case scenario	Inputs	outcomes	Remark
04	Check the histogram and outcomes for 2024/06/17(new csv file)	17 06 2024	Print the outcomes and display the histogram	pass

Test case 04:

```

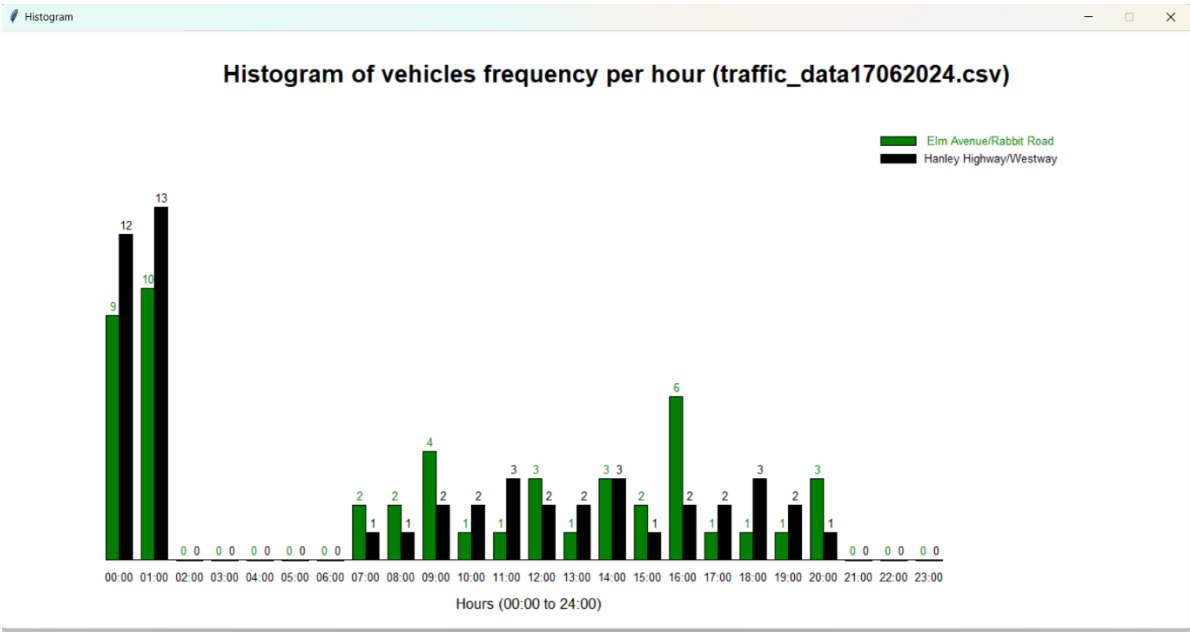
Please enter the day of the survey in the format DD: 17
Please enter the month of the survey in the format MM: 06
Please enter the year of the survey in the format YYYY: 2024

---File 'traffic_data17062024.csv' loaded successfully---

*****
data file selected is traffic_data17062024.csv
The total number of vehicles recorded for this day is 102
The total number of Trucks recorded for this day is 8
The total number of Electric vehicles recorded for this day is 39
The total number of two wheeled vehicles for this date is 36
The total number of busses leaving Elm Avenue/Rabbit Road junction heading north is 1
The total number of vehicles passing through both junctions without turning left or right is 37
The percentage of total vehicles recorded that are Trucks for this date is 8%
The average number Bicycles per hour for this date is 1
The total number of vehicles recorded as over the speed limit for this date is 21
The total number of vehicles recorded through only Elm Avenue/Rabbit Road junction is 50
The total number of vehicles recorded through only Hanley Highway/Westway junction is 52
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters
The highest number of vehicles in an hour on Hanley Highway/Westway is 13
The most vehicles through Hanley Highway/Westway were recorded Between 01:00 and 02:00
The number of hours of rain for this date is 2

Processed data has been successfully saved to 'results.txt'
click once on histogram if you want to select a data file for a different date.
*****

```



3.check about the continue user input after histogram

Test case	Test case scenario	Inputs	outcomes	Remark
05	Check about the continue user input after histogram	Click once on the histogram	ask from the user about want to continue or not	pass

Test case 05:

```
*****
Do you want to select a data file for a different date? (Y/N): |
```

4.handle the error when user close the histogram before mouse click

Test case	Test case scenario	Inputs	outcomes	Remark
06	handle the error when user close the histogram before mouse click	Close the histogram window before mouse click	Print the error message and ask want to select another data file	pass

Test case 06:

```
window closed before mouse click.
Do you want to select a data file for a different date? (Y/N): |
```